Cloud computing – a.y. 2023/2024

Final Project - Analyze letter frequency through Hadoop

Consider a Hadoop cluster installed in **fully distributed** mode (as described in the installation tutorial that I showed). Given text documents as input, the goal of this project is to define a MapReduce algorithm and analyze such documents through Hadoop to extract meaningful results on letter frequency in one or more languages, i.e., how frequent is each letter in one or more languages. In this analysis, consider only standard characters from the alphabet of that language and **ignore any special character**. Assuming English as example, consider only the 26 standard characters and ignore any special character, such as: -_,.!?'$@"#.

In this project, perform experiments by using several text input datasets **of your choice**, spanning **from few KBs to some GBs**. The objective is to reach several InputSplits (and therefore several Mapper tasks) with the biggest input datasets.

Students must demonstrate full knowledge of MapReduce and Hadoop and include the **following features** in their project:

- Combiner and In-Mapper combining
- `Setup()` and `Cleanup()` methods
- Run experiments with different numbers of Reducer tasks, starting from one

Experimental results may be in the form of charts and/or tables and must have a two-fold nature:

- Report insights on letter frequency in languages, for example focusing on a specific language, or comparing different languages, or the same language in different historical periods
- Report statistics on MapReduce execution, e.g., execution time, memory consumption, number of InputSplits, impact of In-Mapper combining

Optionally, students may compare performance of Hadoop execution vs. Python execution on a local machine.

Students must write a **brief** (few pages) **report** describing their algorithm design through pseudocode, the experimental settings and the obtained results.

FAQs on the Cloud computing project and final exam

1. **When and how to upload the final project?**
   - The final project (code + report) must be uploaded through the **Google form** at: https://forms.gle/ycfMLuy5qaMsgG1q8
   - Each group has to submit their project as a .zip file named "Group{Name}_Session{DD-MM-YYYY}.zip"
   - Please, do not include the input datasets in the .zip file
   - Project upload must be performed one week before the official session day, at the latest. For example, if the session is scheduled on the 12/06/2024, students have time to upload their project by 04/06/2024 (included). **Projects that are uploaded after the deadline will be considered for the next session**
   - Once the project is uploaded, **no further modifications** are allowed to it

2. **What about the project discussion?**
   - Project discussion will take place on the official session day or few days before, depending on the amount of projects submitted in that session. In the second case, I will inform you on the exact date and time
   - **All group members** must be present that day and discuss the project as part of their exam
   - Project discussion will consist of **two parts**:
     - For the first part (duration 15 minutes – hard time), students have to prepare a presentation (either through a separate PowerPoint or directly commenting the project report) of the MapReduce pseudocode and related design choices, comment the experimental settings and obtained results, and show an execution demo of their work on the cluster
     - The second part (no duration pre-established) will be a detailed discussion of Java code
   - All the group members must show **complete knowledge** of each aspect of the project. If a student does not demonstrate active involvement in the different parts of the project, that student will have to develop the project assigned to single students and discuss it in another exam session
   - Discussion of the project is in English if at least one group member is a non-Italian speaker. Otherwise, the group can choose to discuss their project either in English or in Italian

3. **What about the oral exam?**
   - The overall exam consists of Hadoop project discussion and the oral exams for my part and Prof. Vallati's one
   - Students can take the different parts of the exam in **different sessions and in any order**
   - Each part of the exam will have a **validity of one year**. After that deadline, that part must be repeated by the student. For example, if you discuss my oral part on 12/06/2024, this will be valid until 11/06/2025
   - Oral exams will take place on the **official session days** only
   - **Groups do not stand** for oral exams: members of the same group can take oral exams in different sessions
   - Students can choose to take their oral exam either in English or in Italian